Wikidata for the people of Africa

—

Laurette Marais, GFSS 2025

Stellenbosch UNIVERSITY IYUNIVESITHI UNIVERSITEIT

CSIR Touching lives through innovation

Overview

- Introduction
- Basic idea
- Wikidata for the (isiZulu) people of Africa
- Work plan
- Procure isiZulu terminology/translations
- An application grammar for descriptions
- Generating true descriptions
- Conclusion

Introduction

Wikidata is “a free and open knowledge base that can be read and edited by both humans and machines”.

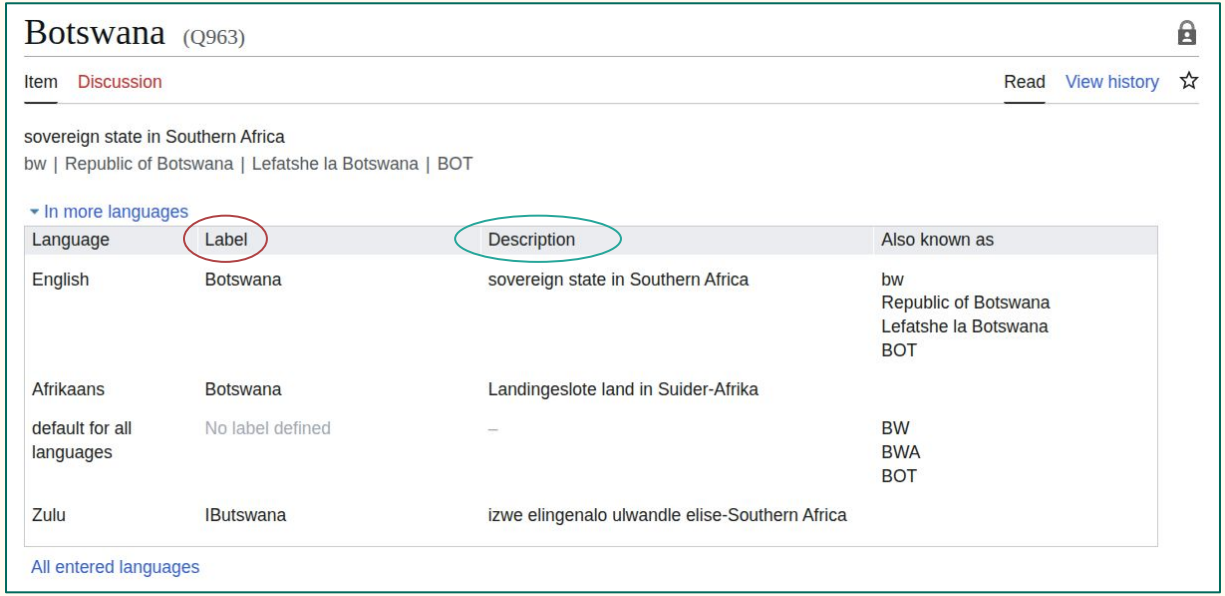
- Editing, consuming, browsing, and reusing the data is fully multilingual
- Editing in any language is possible and encouraged.

Entities are related via *properties*.

Eg. *Botswana* is an *instance of* a *landlocked country*.

Introduction

Entities (and properties) have natural language *labels* and *descriptions*.



Botswana (Q963)

Item Discussion Read View history

sovereign state in Southern Africa

bw | Republic of Botswana | Lefatshe la Botswana | BOT

In more languages

Language	Label	Description	Also known as
English	Botswana	sovereign state in Southern Africa	bw Republic of Botswana Lefatshe la Botswana BOT
Afrikaans	Botswana	Landingeslote land in Suider-Afrika	
default for all languages	No label defined	–	BW BWA BOT
Zulu	IButswana	izwe elingenalo ulwandle elise-Southern Africa	

All entered languages

Introduction

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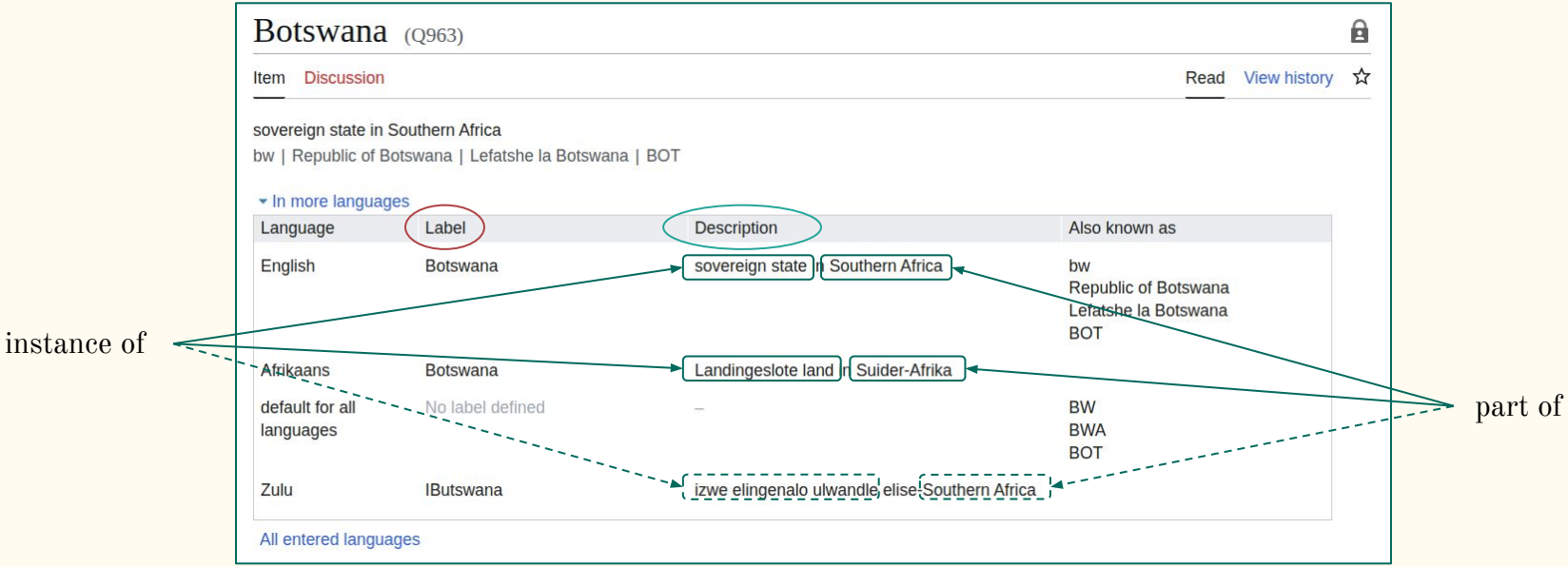


Basic idea

Statements

instance of	sovereign state 0 references
	landlocked country 1 reference
	country 0 references

part of	Southern Africa 1 reference

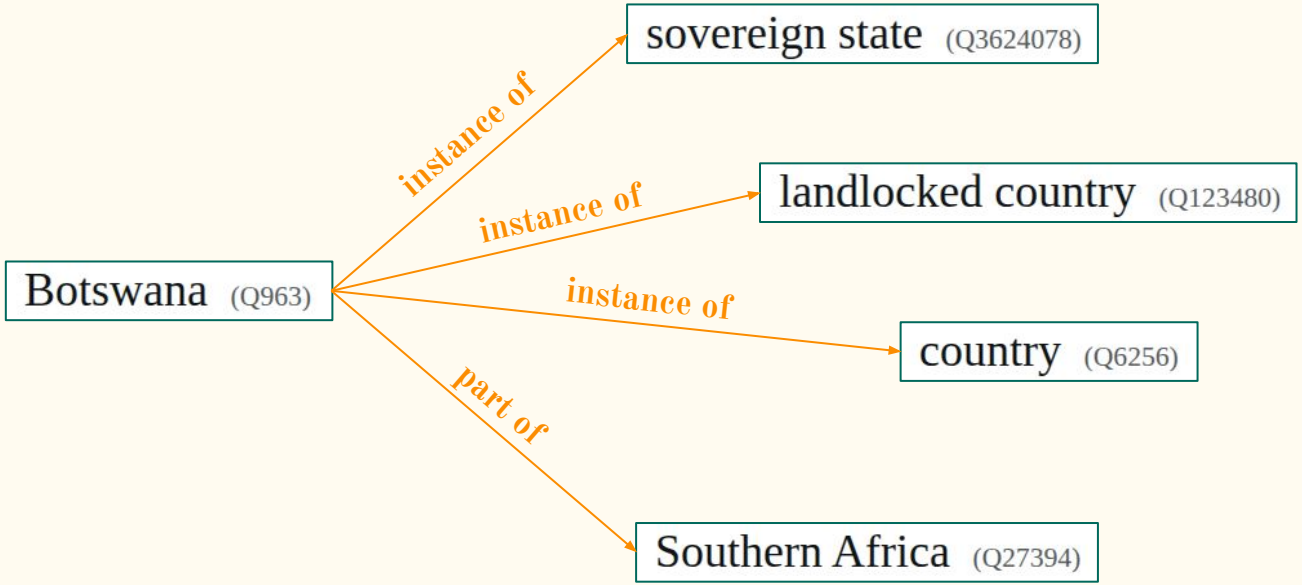
sovereign state (Q3624078)

landlocked country (Q123480)

country (Q6256)

Southern Africa (Q27394)

Basic idea


Botswana (Q963)
instance of
sovereign state (Q3624078)
instance of
landlocked country (Q123480)
instance of
country (Q6256)
part of
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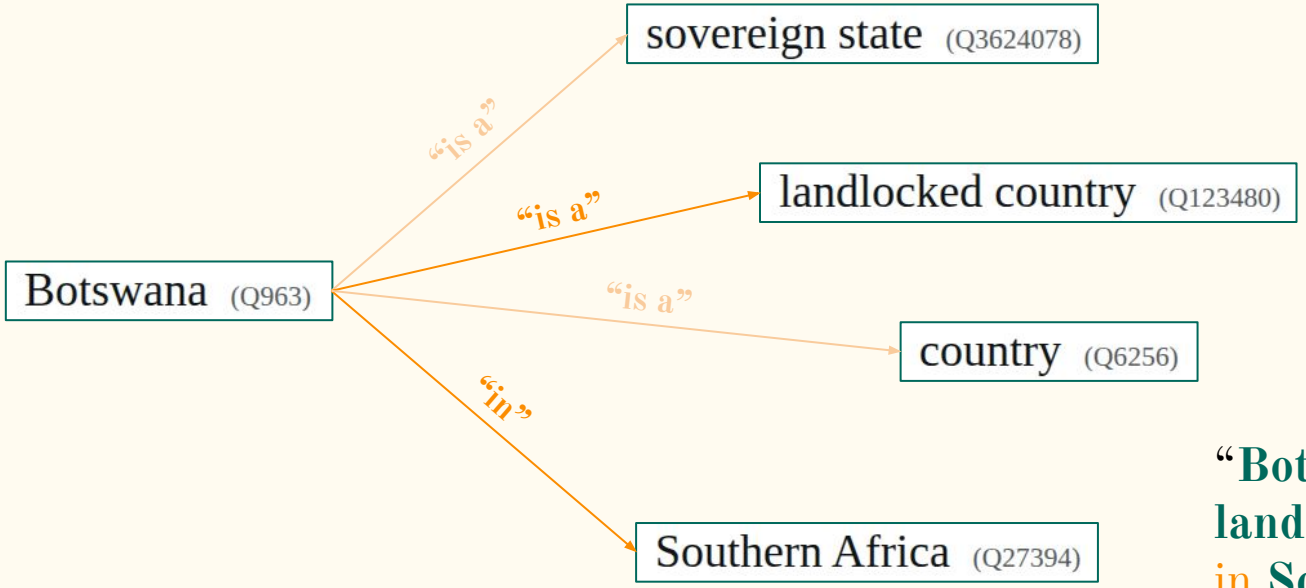
Basic idea



"**Botswana** is a **landlocked country** in **Southern Africa**"

Wikidata for the (isiZulu) people of Africa

Kiribati dollar (Q1049963)

Item Discussion Read View history ☆

currency edit

In more languages

Language	Label	Description	Also known as
English	Kiribati dollar	currency	
Afrikaans	No label defined	No description defined	
default for all languages	No label defined	–	
Zulu	idola lase-Kiribati	uhlobo lwemali olusemthethweni wase-Kiribati	

country

Wikidata for the (isiZulu) people of Africa

Goal - Contribute 774 labels and 1005 descriptions to the Wikidata knowledge graph for isiZulu within the geopolitical domain.

Deliverable 1 components	**# English**	**# isiZulu**	**Estimated minimum contribution**
All countries and relevant subclasses - labels	254	135	119 isiZulu labels
All countries and relevant subclasses - labels and descriptions	252	6	246 isiZulu descriptions
Capital cities of countries and relevant subclasses - labels	248	57	191 isiZulu labels
Capital cities of countries and relevant subclasses - labels and descriptions	248	0	248 isiZulu descriptions
Languages with more than a million speakers - labels	350	49	301 isiZulu labels
Languages with more than a million speakers - labels and descriptions	350	6	344 isiZulu descriptions
Currencies of countries and relevant subclasses - labels	167	4	163 isiZulu labels
Currencies of countries and relevant subclasses - labels and descriptions	166	0	167 isiZulu descriptions
Total labels			**774 isiZulu labels**
Total descriptions			**1005 isiZulu descriptions**

Work plan

1. Identify relevant entities (instances of countries, capital cities, languages, currencies)

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5. Procure isiZulu terminology/translations for above lists.

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4. Compile a list of English phrases that represent typical descriptions of the original entities.
5. Procure isiZulu terminology/translations for above lists.
6. Develop application grammar that supports typical descriptions.
7. Develop an NLG script that queries the knowledge graph and determines trees that could be used to generate descriptions.

Procure isiZulu terminology/translations

```
# All capital cities of an instance of a country or a subclass thereof
SELECT DISTINCT ?city ?cityLabel_en ?cityDescription_en WHERE {
  ?country wdt:P31/wdt:P279* wd:Q6256 .
  ?country wdt:P36 ?city .
  ?city rdfs:label ?cityLabel_en . FILTER(lang(?cityLabel_en)='en') .
  ?city schema:description ?cityDescription_en . FILTER(lang(?cityDescription_en)='en') .
}
ORDER BY ?cityLabel_en
```

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}
ORDER BY ?cityLabel_en
```

city	cityLabel_en	cityDescription_en
wd:Q1519	Abu Dhabi	capital city of the United Arab Emirates
wd:Q3787	Abuja	capital city of Nigeria
wd:Q3761	Accra	capital city of Ghana
wd:Q3624	Addis Ababa	capital city of Ethiopia
wd:Q131694	Aden	port city and temporary capital of Yemen
wd:Q193809	Aigues-Mortes	commune in Gard, France
wd:Q289618	Al-Mansuriya	place in Tunisia, Ifriqiya
wd:Q3561	Algiers	capital city of Algeria

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No existing isiZulu terminology!

Procure isiZulu terminology/translations

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isiZulu translation
I-Abu Dhabi
I-Abuja
I-Accra
I-Addis Ababa
I-Aden
I-Al-Mansuriya
I-Algiers

An application grammar for descriptions

```
-- currency names nouns
oper
    Afghani_N : N = mkN "-Afghani" ;
    Albanian_lek_CN = possCN (mkN "-lek" C5_6) (mkNP (mkLN "-Albania")) ;
    Algerian_dinar_CN : CN = possCN (mkN "-dinar" C5_6) (mkNP (mkLN "-Algeria")) ;
    Argentine_convertible_peso_CN : CN = possCN (mkN "-peso" C5_6) (mkNP (mkLN "-Argentinia")) ;
    Armenian_dram_CN = possCN (mkN "-dram" C5_6) (mkNP (mkLN "-Armenia")) ;
    Aruban_florin_CN = possCN (mkN "-florin" C5_6) (mkNP (mkLN "-Aruba")) ;
    Australian_dollar_CN : CN = possCN (mkN "dola" C5_6) (mkNP (mkLN "-Australia")) ;
    Azerbaijani_manat_CN : CN = possCN (mkN "-manat" C5_6) (mkNP (mkLN "-Azerbaijan")) ;
    Bahamian_dollar_CN : CN = possCN (mkN "dola" C5_6) (mkNP (mkLN "-Bahamas")) ;
    Bahraini_dinar_CN : CN = possCN (mkN "-dinar" C5_6) (mkNP (mkLN "-Bahrain")) ;
    Barbadian_dollar_CN : CN = possCN (mkN "dola" C5_6) (mkNP (mkLN "-Barbados")) ;
    Bangladeshi_taka_CN : CN = possCN (mkN "-taka" C5_6) (mkNP (mkLN "-Bangladesh")) ;
    Belarusian_ruble_CN : CN = possCN (mkN "-ruble" C5_6) (mkNP (mkLN "-Belarusia")) ;
    Belize_dollar_CN : CN = possCN (mkN "dola" C5_6) (mkNP (mkLN "-Belize")) ;
    Bhutanese_ngultrum_CN = possCN (mkN "-ngultrum") (mkNP (mkLN "-Bhutan")) ;
    Boliviano_N : N = mkN "-boliviano" ;
    Botswana_pula_CN : CN = possCN (mkN "-pula" C5_6) (mkNP (mkLN "-Botswana")) ;
```

Terminology captured in resource module `WikidataLexiconZul.gf`.

An application grammar for descriptions

```
lin
    Q199471_Afghani = mkNP Afghani_N ;
    Q125999_Albanian_lek = mkNP Albanian_lek_CN ;
    Q199674_Algerian_dinar = mkNP Algerian_dinar_CN ;
    Q200337_Angolan_kwanza = mkNP Kwanza_N ;
    Q199578_Argentine_convertible_peso = mkNP Argentine_convertible_peso_CN ;
    Q130498_Armenian_dram = mkNP Armenian_dram_CN ;
    Q232270_Aruban_florin = mkNP Aruban_florin_CN ;
    Q259502_Australian_dollar = mkNP Australian_dollar_CN ;
    Q483725_Azerbaijani_manat = mkNP Azerbaijani_manat_CN ;
    Q194339_Bahamian_dollar = mkNP Bahamian_dollar_CN ;
    Q201871_Bahraini_dinar = mkNP Bahraini_dinar_CN ;
    Q194453_Bangladeshi_taka = mkNP Bangladeshi_taka_CN ;
    Q194351_Barbadian_dollar = mkNP Barbadian_dollar_CN ;
    Q21531507_Belarusian_ruble = mkNP Belarusian_ruble_CN ;
    Q275112_Belize_dollar = mkNP Belize_dollar_CN ;
    Q201799_Bhutanese_ngultrum = mkNP Bhutanese_ngultrum_CN ;
    Q200737_Boliviano = mkNP Boliviano_N ;
    Q186794_Botswana_pula = mkNP Botswana_pula_CN ;
    Q173117_Brazilian_real = mkNP Brazilian_real_CN ;
    Q206319_Brunei_dollar = mkNP Brunei_dollar_CN ;
    Q172540_Bulgarian_lev = mkNP Bulgarian_lev_CN ;
    Q238007_Burundian_franc = mkNP Burundian_franc_CN ;
    Q204737_Cambodian_riel = mkNP Cambodian_riel_CN ;
    Q1104069_Canadian_dollar = mkNP Canadian_dollar_CN ;
    Q4591_Cape_Verdean_escudo = mkNP Cape_Verdean_escudo_CN ;
    Q847739_Central_African_CFA_franc = mkNP Central_African_CFA_franc_CN ;
    Q39099_Chinese_renminbi = mkNP Renminbi_N ;
```

Linearisation functions for terms…

Procure isiZulu terminology/translations

Elicitation of typical descriptions from translators

official language of Botswana	ulimi olusemthethweni lase-Botswana
language spoken in Botswana	ulimi olukhulunywa e-Botswana
language spoken in Botswana and Namibia	ulimi olukhulunywa e-Botswana nase-Namibia
language spoken in 7 countries	ulimi olukhulunywa emazweni ayisi-7
language with 16000 speakers	ulimi olunezikhulumi ezingu-16000
language spoken in 7 countries and the official language of Kenya	ulimi olukhulunywa emazweni ayisi-7 futhi ulimi olusemthethweni lwase-Kenya
language spoken in 3 countries and an official language of South Africa	ulimi olukhulunywa emazweni ama-3 futhi ulimi olusemthethweni lwaweNingizimu Afrika
language with 16000 speakers, spoken in 2 countries	ulimi olunezikhulumi ezingu-16000, olukhulunywa emazweni ama-2

An application grammar for descriptions

```
-- verbs
is_spoken_V : V = mkV "khulunyw" ;

-- adjectives
big_A : A = mkA "khulu" ;
one_A : A = mkA "nye" ;

-- nouns
country_N : N = mkN "zwe" C5_6 ;
island_N : N = mkN "qhingi" C7_8 ;
territory_N : N = mkN "ndawo" C9_10 ;
state_N : N = mkN "buso" C3_4 ;
union_N : N = mkN "nyunyana" C9_10 ;
continent_N : N = mkN "zwekazi" C5_6 ;
municipality_N : N = mkN "masipala" C1a_2a ;

capital_city_N : N = mkN "nhlokodolobha" C9_10 ;
city_N : N = mkN "dolobha" C5_6 ;
language_N : N = mkN "limi" C11_10 ;
speaker_N : N = mkN "khulumi" C7_8 ;
ocean_N : N = mkN "lwandle" C11_10 ;
law_N : N = mkN "thetho" C3_4 ;
type_N : N = mkN "hlobo" C11_10 ;
money_N : N = mkN "mali" C5_6 ;
```

Capture some general vocabulary in `WikidataLexiconZul.gf`.

An application grammar for descriptions

Range	Head	Fully normalised example (abantu...)	Literal translation (people...)	Example (abantu...)
2-5	Adjective	ababili	who are two	aba-2
6-9	Noun (class 7)	abayisithupha	who are six	abayisi-6
10-19	Noun (class 5)	abayishumi nambili	who are ten and two	abayi-12
20-99	Noun (class 6)	abangamashumi amabili	who are tens that are two	abangama-20
100-199	Noun (class 5)	abayikhulu namashumi ayisikhombisa	who are a hundred and tens that are seven	abayi-170
200-999	Noun (class 6)	abangamakhulu amahlanu	who are hundreds that are five	abangama-500
1000-1999	Noun (class 9)	abayinkulungwane namashumi amabili	who are a thousand and tens that are two	abayin-1020
2000-999 999	Noun (class 10)	abayizinkulungwane ezine	who are thousands that are four	abayizin-4000

An application grammar for descriptions

```
abstract SymbolSBantu = Cat, Symbol ** {

    fun
        SymbN : Symb -> N ;
        SymbLoanPlaceName : Symb -> N ;
        SymbA : Symb -> A ;

        SymbN_sub10 : Symb -> N ;
        SymbN_10 : Symb -> N ;
        SymbN_100 : Symb -> N ;
        SymbN_1000 : Symb -> N ;
        SymbN_1000_000 : Symb -> N ;
}
```

Add handling of strings as different kinds of numbers in isiZulu to the resource grammar.

```
SymbN_sub10 symb = mkSymbNumber symb.s C7_8 ;
SymbN_10 symb = mkSymbNumber symb.s C5_6 ;
SymbN_100 symb = mkSymbNumber symb.s C5_6 ;
SymbN_1000 symb = mkSymbNumber symb.s C9_10 ;
SymbN_1000_000 symb = mkSymbNumber symb.s C7_8 ;
```

An application grammar for descriptions

```
abstract Descriptions =
    Countries,
    Languages,
    Cities,
    Currencies,
    SymbolSBantu ** {

    flags
        startcat = Description ;

    cat
        Description ;
        CityFeature ;
        CurrencyFeature ;
        LanguageFeature ;
        NumberMod ;
        SymbNumber ;
        AdminTerritory ;
```

```
fun
    -- countries
    CountryKindDescription : CountryKind -> Description ;
    CountryRegionDescription : CountryRegion -> Description ;
    FullCountryDescription : CountryKind -> CountryRegion -> Description ;

    -- cities
    CityDescription : Country -> Description ;
    CityFeatureDescription : CityFeature -> Country -> Description ;

    -- languages
    LanguageFeatureDescription : LanguageFeature -> Description ;

    -- currencies
    CurrencyFeatureDescription : CurrencyFeature -> Description ;

    -- city features
    LargestCity : CityFeature ;

    -- language features
    OfficialLanguage : Country -> LanguageFeature ;
    SpokenInCountry: Country -> LanguageFeature ;
    SpokenInTwoCountries : Country -> Country -> LanguageFeature ;
    SpokenCountriesNumber : NumberMod -> LanguageFeature ;
    NumberOfSpeakers : NumberMod -> LanguageFeature ;
    SpokenCountriesAndTheOfficial : Country -> NumberMod -> LanguageFeature ;
    SpokenCountriesNumberOfSpeakers : NumberMod -> NumberMod -> LanguageFeature ;

    -- currency features
    OfficialCurrency : Country -> CurrencyFeature ;
    CentralBankFeature : CentralBank -> CurrencyFeature ;
```

Abstract syntax for geopolitical descriptions.

An application grammar for descriptions

```
concrete DescriptionsZul of Descriptions =
    CountriesZul,
    CurrenciesZul,
    LanguagesZul,
    CitiesZul,
    SymbolSBantuZul **
    open
    SyntaxSBantuZul,
    ParadigmsZul,
    Prelude,
    ResZul,
    WikiDataLexiconZul in {

    lincat
        Description = NP ;
        CityFeature = NP ;
        LanguageFeature = NP ;
        CurrencyFeature = NP ;
        NumberMod = VP ;
        SymbNumber = Symb ;
        AdminTerritory = N ;
```

```
lin
    -- countries
    CountryKindDescription ck = ck ;
    CountryRegionDescription cr = mkNP (mkNP country_N) (mkRS (mkVP (mkLoc (locativisedNoun cr)))) ;
    FullCountryDescription ck cr = mkNP ck (mkRS (mkVP (mkLoc (locativisedNoun cr)))) ;

    -- cities
    CityDescription c = mkNP (possCN capital_city_N c) ;
    CityFeatureDescription cf c = mkNP cf together_with_ConjN (mkNP (possCN capital_city_N c)) ;

    -- languages
    LanguageFeatureDescription f = f ;

    -- currencies
    CurrencyFeatureDescription f = f ;

    -- city features
    LargestCity = mkNP (mkNP city_N) (mkRS (mkVP (mkAP big_A))) ;

    -- language features
    OfficialLanguage c = mkNP (mkNP language_N) (mkRS (mkVP (mkLoc (locativisedNoun (mkNP (possCN law_N
    c))))))) ;
    SpokenInCountry c = mkNP (mkNP language_N) (mkRS (mkVP (mkVP is_spoken_V) (SyntaxSBantuZul.mkAdv c))) ;
    SpokenInTwoCountries c1 c2 = mkNP (mkNP language_N) (mkRS (mkVP (mkVP is_spoken_V) (SyntaxSBantuZul.
    mkAdv (mkNP c1 with_ConjN c2)))) ;
    SpokenCountriesNumber nummod = mkNP (mkNP language_N) (mkRS (mkVP (mkVP is_spoken_V) (SyntaxSBantuZul.
    mkAdv (mkNP (mkNP aPl_Det country_N) (mkRS nummod))))) ;
    NumberOfSpeakers nummod = mkNP (mkNP language_N) (mkRS (assocVP (mkNP (mkNP aPl_Det speaker_N) (mkRS
    nummod)))) ;
    SpokenCountriesAndTheOfficial c nummod = mkNP (mkNP language_N) (mkRS (mkRS (mkVP (mkVP is_spoken_V)
    (SyntaxSBantuZul.mkAdv (mkNP (mkNP aPl_Det country_N) (mkRS nummod))))) and_Conj (mkRS (mkVP (mkLoc
    (locativisedNoun (mkNP (possCN law_N c)))))))) ;
    SpokenCountriesNumberOfSpeakers nums numc = mkNP (mkNP language_N) (mkRS (mkRS (assocVP (mkNP (mkNP
    aPl_Det speaker_N) (mkRS nums)))) and_Conj (mkRS (mkVP (mkVP is_spoken_V) (SyntaxSBantuZul.mkAdv (mkNP
    (mkNP aPl_Det country_N) (mkRS numc)))))) ;

    -- currency features
    OfficialCurrency c = mkNP (mkNP type_N (mkNP money_N)) (mkRS (mkVP (mkLoc (locativisedNoun (mkNP
    (possCN law_N c)))))) ;
    CentralBankFeature cb = mkNP (mkNP type_N (mkNP money_N)) (mkRS (mkVP be_taken_out_V2 cb)) ;
```

Concrete syntax for geopolitical descriptions.

Generating true descriptions

```
# All capital cities of an instance of a country or a subclass thereof
SELECT DISTINCT ?item WHERE {
  ?country wdt:P31/wdt:P279* wd:Q6256 .
  ?country wdt:P36 ?item .
}
```

```
def get_entities(query_filename):
    # get all entities
    with open(query_filename) as query_file:
        QUERY = query_file.read()
    generator = pg.WikidataSPARQLPageGenerator(QUERY, site=WIKIDATA_SITE)

    try:
        entities = list(itertools.islice(generator, QUERY_LENGTH_LIMIT))
    except TypeError as e:
        warning_msg(f"Could not retrieve entities using {query_filename}")
        return

    return entities
```

```
entity_dict = entity.get()
entity_id = entity.getID()
```

Generating true descriptions

```python
def get_capital_city_tree_str(city_id,city_dict,pgf_grammar):

    claims_dict = city_dict['claims']
    country_claims = claims_dict['P17']

    for country_claim in country_claims:
        qualifiers = country_claim.qualifiers
        if not 'P582' in qualifiers.keys(): # if the country claim has no end date
            break
    country_tgt = country_claim.getTarget()
    if country_tgt:
        country_id = country_tgt.getID()
    else:
        return
    if city_id == country_id:
        return # city state, will deal with this as a country
    country_fun = get_function_name_of_entity(country_id,pgf_grammar)

    instance_of_claims = claims_dict["P31"]
    for instance_of_claim in instance_of_claims:
        clm_target = instance_of_claim.getTarget()
        if clm_target.getID() == 'Q51929311': # largest city
            tree_str = f"CityFeatureDescription LargestCity {country_fun}"
        else:
            tree_str = f"CityDescription {country_fun}"

    if not country_fun:
        return

    return tree_str
```

find the city's current country

get the country's id

get the country's fun

use largest city feature if applicable

fall back to country only description

Generating true descriptions

```yaml
Q1335_Montevideo_CapitalCity:
  description_tree: CityDescription Q77_Uruguay
  descriptions:
    zul: inhlokodolobha yase-Uruguay
  gf_id: Q1335_Montevideo
  labels:
    zul: i-Montevideo
  wd_id: Q1335
Q13389_Bashkir_Language:
  description_tree: LanguageFeatureDescription (SpokenCountriesAndTheOfficial Q159_Russia
    (SymbModA (SymbNumb (MkSymb "4"))))
  descriptions:
    zul: ulimi olukhulunywa emazweni ama-4 futhi olusemthethweni wase-Russia
  gf_id: Q13389_Bashkir
  labels:
    zul: isi-Bashkir
  wd_id: Q13389
Q1354_Dhaka_CapitalCity:
  description_tree: CityFeatureDescription LargestCity Q902_Bangladesh
  descriptions:
    zul: idolobha elikhulu kanye nenhlokodolobha yase-Bangladesh
  gf_id: Q1354_Dhaka
  labels:
    zul: i-Dhaka
  wd_id: Q1354
```

Generate descriptions for all targeted entities.

Generating true descriptions

```
Q1335 Lzul "i-Montevideo"
Q1335 Dzul "inhlokodolobha yase-Uruguay"
Q13389 Lzul "isi-Bashkir"
Q13389 Dzul "ulimi olukhulunywa emazweni ama-4 futhi olusemthethweni wase-Russia"
Q1354 Lzul "i-Dhaka"
Q1354 Dzul "idolobha elikhulu kanye nenhlokodolobha yase-Bangladesh"
```

Contribute new labels and descriptions via batch upload to Wikidata.

- (cur | prev) ◯ 14:13, 30 June 2025 WierieR (talk | contribs) . . (112,805 bytes) (+126) . . *(Added [zu] description: ulimi olukhulunywa emazweni ama-4 futhi olusemthethweni wase-Russia, #quickstatements; #temporary_batch_1751292621515)* (thank) (Tag: quickstatements [2.0])
- (cur | prev) ◯ 14:13, 30 June 2025 WierieR (talk | contribs) . . (112,679 bytes) (+70) . . *(Added [zu] label: isi-Bashkir, #quickstatements; #temporary_batch_1751292621515)* (thank) (Tag: quickstatements [2.0])

A different domain?

```
LocalMunicipalityDescription: AdminTerritory -> Country -> Description ;
SymbAdminTerritory : Symb -> AdminTerritory ;
```

abstract functions

```
LocalMunicipalityDescription territory country =
    mkNP
        (mkNP (mkCN municipality_N (mkNP territory_N)))
        (SyntaxSBantuZul.mkAdv (locativisedNoun (mkNP (possCN territory country)))) ;

SymbAdminTerritory s = loanPlaceNameN s ;
```

concrete functions

```
def get_municipality_tree_str(municipality_id,municipality_dict,pgf_grammar):

    claims_dict = municipality_dict["claims"]

    # ...
```

natural language generation

A different domain?

```
Q1255574_LocalMunicipality:
  description_tree: LocalMunicipalityDescription (SymbAdminTerritory (MkSymb "Cape
    Winelands District Municipality")) Q258_South_Africa
  descriptions:
    zul_new: umasipala wendawo e-Cape Winelands District Municipality laseNingizimu
      Afrika
    zul_old: null
  gf_id: Q1255574
  labels:
    eng: Drakenstein Local Municipality
    zul_new: '[Q1255574]'
    zul_old: Drakenstein Local Municipality
  wd_id: Q1255574
Q1298937_LocalMunicipality:
  description_tree: LocalMunicipalityDescription (SymbAdminTerritory (MkSymb "UmGungundlovu"))
    Q258_South_Africa
  descriptions:
    zul_new: umasipala wendawo e-UmGungundlovu laseNingizimu Afrika
    zul_old: null
  gf_id: Q1298937
  labels:
    eng: Mpofana Local Municipality
    zul_new: '[Q1298937]'
    zul_old: Mpofana Local Municipality
```

Conclusion

- Severe lack of terminology and expertise to develop it
- This “simplifies” things for contributing labels and descriptions to Wikidata
- Contributing a new language would require
  - Adding a new language to the GF application grammar
  - Likely that other Bantu languages have the same mix of native vocabulary and loaned terminology
  - That’s probably it…
- Targeting a new domain could be done via
  - Similar elicitation of typical description structures
  - Relying more heavily on being able to “simply” loan English terminology
  - However, targeting a domain that *contains isiZulu names and terms* would be more complex!

https://github.com/LauretteM/w4pa